Milestone 1: Concept Drafts

Human, Computer, and AI Interaction. Term Project 2026

Track 9: Health Information Navigator. Principle: Safety and Ethics

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Repository: https://github.com/KareemHerish/HCAI.git

1. Project idea for approval

Working name: Health Information Navigator.

What it does: A web chatbot that gives general, non-diagnostic health information from trusted public sources (MedlinePlus, CDC, WHO). It never diagnoses, never gives medication doses, and hands over to real people when a user may be at risk.

Target users: the general public in Egypt. Adults of any age and any technical level who look for general, trustworthy health information. Not limited to a specific group.

Principle in the AI pipeline (not only on screen):

- A red-flag filter runs before retrieval (chest pain, breathing difficulty, self-harm language) and stops the normal answer.
- Rules block diagnosis and dosing requests.
- A minimum similarity threshold: if evidence is weak, the bot says it is not sure and suggests a pharmacist or doctor.
- An escalation banner shows emergency numbers (Egypt ambulance: 123) and how to reach a real person.

Knowledge source: public pages from MedlinePlus, CDC and WHO. We will check each license in Week 3 and keep a local copy of the selected passages.

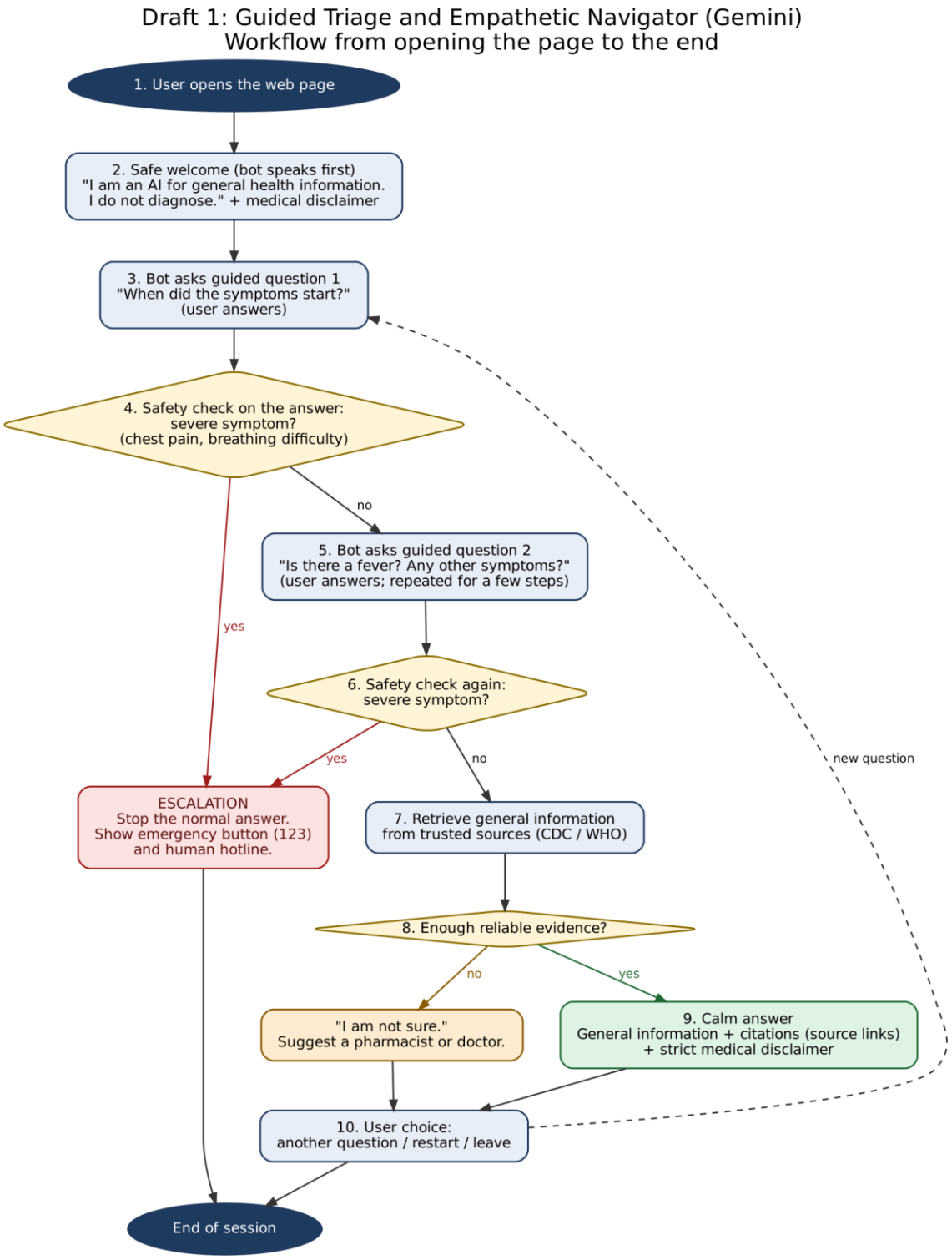
Own-track proposal: not needed. We use the existing Track 9.

2. Two concept drafts side by side

Draft 1 was generated with Gemini. Draft 2 was generated with Claude. The exact prompts are in Section 4.

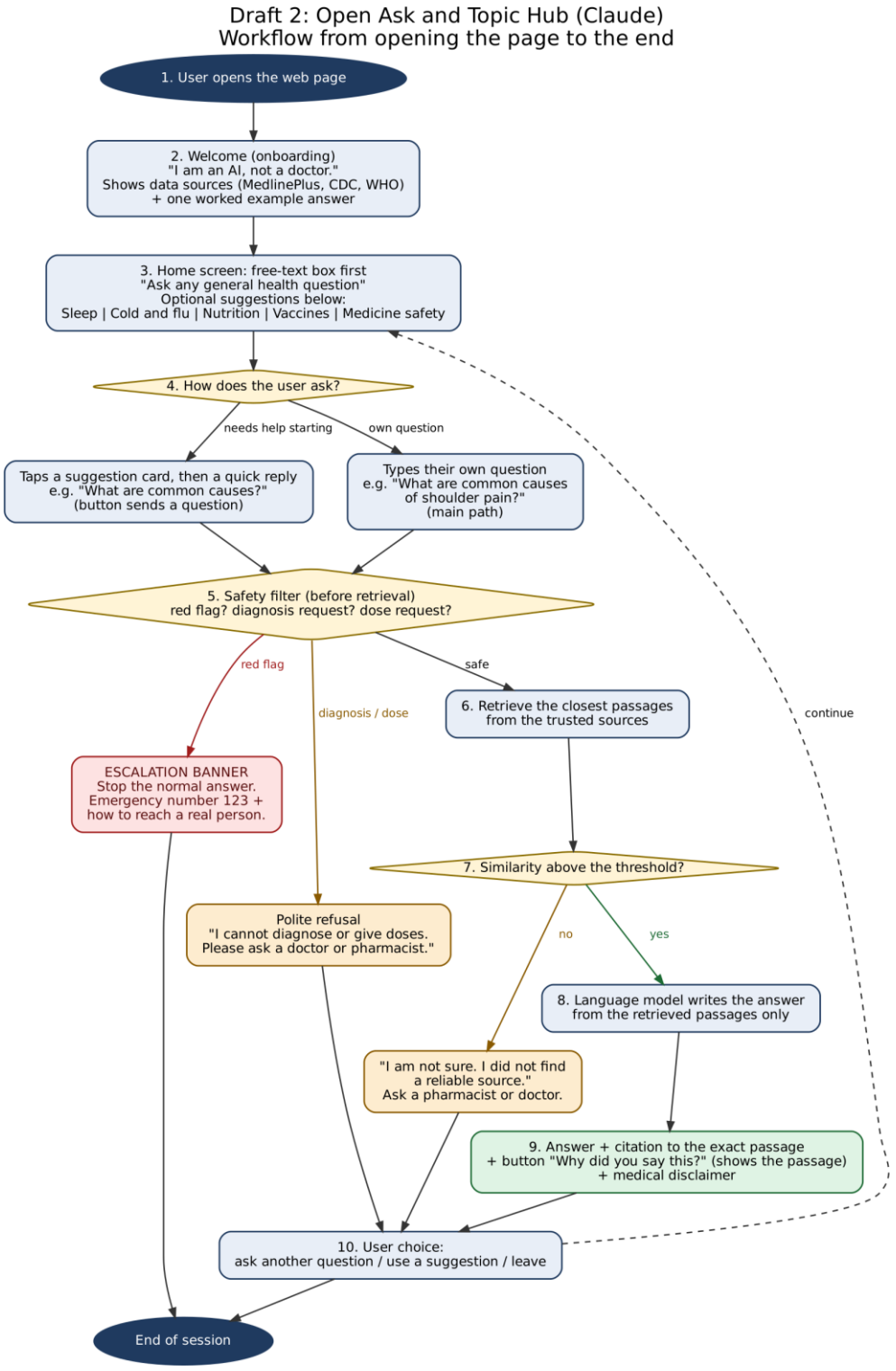
Item	Draft 1: Guided Triage and Empathetic Navigator (Gemini)	Draft 2: Open Ask and Topic Hub (Claude)
Core idea	A guided, step-by-step flow. The bot asks short questions about symptoms, screens for emergencies, and shows trusted sources under each answer.	The user leads. They start with their own question in a free-text box. Topic cards and quick replies sit below as optional suggestions for users who do not know how to start.
Step 1	Safe welcome: says it is an AI for general health information and gives no diagnosis.	Welcome: says it is an AI, not a doctor, lists its data sources, and shows one worked example.
Step 2	Symptom collection with multi-step guided questions (when did it start? is there a fever?).	User types any general health question (e.g. "What are common causes of shoulder pain?"), or taps a topic card and a quick reply, which sends a question.
Step 3	Calm answer from the trusted database with precise citations.	Every question, typed or sent by a button, is checked by the safety filter before retrieval.
Step 4	Escalation: serious symptoms stop the normal answer and show an emergency button or a human hotline.	Answer with a citation to the exact passage and a "Why did you say this?" button that shows it.
Escalation	Rules for severe symptoms (chest pain, breathing difficulty).	Red flags (chest pain, breathing difficulty, self-harm) stop the answer and show a banner with emergency contacts.
Weak evidence	Not defined in this draft. Added in the final design (minimum similarity threshold).	If similarity is low: "I am not sure. I did not find a reliable source. Ask a pharmacist or doctor."

2.1 Draft 1 workflow (Guided Triage)



The bot speaks first and leads with guided questions. A safety check runs after each answer, and any severe symptom goes straight to escalation.

2.2 Draft 2 workflow (Open Ask and Topic Hub)



The user asks first, by typing or by tapping a suggestion. Every question passes the safety filter before retrieval, then the answer is written from the retrieved passages only.

3. Comparison and chosen direction

Aspect	Draft 1	Draft 2
Strengths	Clear escalation rules. Guided questions reduce ambiguous input for users who cannot describe their problem.	The user asks about their real concern, so no irrelevant questions. Suggestions help novices and buttons help accessibility (keyboard, motor, cognitive load).
Weaknesses	Sequential symptom questions come close to diagnosis, which the project forbids. The fixed script can be irrelevant (asking about fever when the problem is a sore shoulder). Many paths make it hard to test.	Free text can go out of scope or touch unsafe topics, so it needs a strong safety filter. A blank box can intimidate some users, so the suggestions are needed.
Testing the 30 prompts	Complex: many branches.	Simpler: one question gives one answer. Button questions are fixed, and typed questions are tested as separate groups.

Chosen direction: a hybrid built on Draft 2.

- From Draft 2: the free-text-first input (the user asks first), optional topic suggestions and quick replies, and the three-stage onboarding.
- From Draft 1: the escalation rules and the clear medical disclaimer.
- Removed: the bot-led symptom questions, so the bot stays general information and does not assess a person's case.

4. AI Prompt Log (Appendix A)

#	Tool	Exact prompt	What it gave	What we changed and why
1	Gemini	Act as a UX designer for Track 9 (Health Information Navigator). Design a core interaction flow focusing on safety guardrails, public health dataset retrieval (CDC/WHO), and strict non-diagnostic rules.	A detailed conversational flow with emergency escalation steps and source citation guidelines (Draft 1).	Kept the escalation rules and the disclaimer. Rejected the symptom-collection stage because it is close to diagnosis.
2	Claude	The message beginning "Milestone 1, what each team show:" with the check-in requirements and the pasted Gemini output, ending with "جيه دورك". Full verbatim text in A.1 below.	Draft 2 (topic cards and quick replies), a comparison with Draft 1, a hybrid direction, and an idea for approval.	Dropped detailed symptom collection and merged Gemini's escalation rules. The student user group it added was removed in entry 3.
3	Claude	انا مقولتش ان هو مخصص لل students هو عام	Acknowledged that the student group was added by the AI, not by the team. Changed the target users to the general public.	Target users are now the general public in Egypt.
4	Claude	طب ما تجيب درافت غير درافت ٢ ده كده	A different Draft 2 concept: "Open Ask", where the user starts with their own question, with a confidence badge and an evidence panel.	Kept the free-text-first idea. The confidence badge and evidence panel are not in the Milestone 1 file.
5	Claude	تمام عدل بس كده درافت واحد مازال البوت هو الي بيسأل طب افرض البوت سال اول سؤال انت حرارتك عاليه ؟ و اصلا اليوزر كتفه الي واجعه	Agreed that fixed bot questions can be irrelevant to the user's problem. Rewrote Draft 2 (free-text first, optional topic suggestions) and redrew its workflow diagram.	Added the irrelevant-question weakness to Draft 1. Draft 2 now starts with the user's own question.

A.1 Full verbatim text of prompt 2 (Claude)

Milestone 1, what each team show:

1. Two concept drafts side by side. Use two different AI tools to draft two concepts for the chatbot's core interaction flow.
2. A short comparison of the strengths and weaknesses of the two drafts, and which direction they chose.
3. Prompt log entries. These are the exact prompts, the tool name, what it gave, and what the team changed and why (Appendix A).
4. The idea for approval. It must be unique < since two teams can share a track but not the same user group and idea>

*If a team is proposing its own track, a one-paragraph proposal.

Rules for the check-in:

- Teams must open the file or run the demo, not just describe it.

- Any team member can be asked to explain any output.

ده الي مطلوب الاسبوع ده و احنا اخترنا تراك ٩ و كمان استخدمت جيميناي عشان هو مطلوب نستخدم ٢ ai tool
جيه دورك
ده ناتج جيميناي

Concept Draft 1: "The Guided Triage & Empathetic Navigator" (المفهوم الأول)

التركيز الأساسي: يعتمد على نظام تدرج إرشادي ذكي يبدأ بسؤال المستخدم عن أعراضه بطريقة مبسطة، مع فرز فوري للحالات الطارئة جداً لتوجيهها للطوارئ، مع إبراز مصادر موثوقة (مثل CDC أو WHO) أسفل كل إجابة لضمان Transparency، وإخلاء مسؤولية طبي صارم وواضح في الواجهة.

تدفق التفاعل (Core Interaction Flow):

الترحيب الآمن: يرحب البوت بالمستخدم، يعرفه بأنه ذكاء اصطناعي مخصص للصحة العامة، ويؤكد فوراً أنه لا يقدم تشخيصاً.

جمع الأعراض: يطرح أسئلة تفاعلية قصيرة (Multi-step guided questions) لفهم المشكلة (مثل: متى بدأت الأعراض؟ هل تصاحبها حرارة؟).

الاستجابة الملطفة والمستندة للمصادر: يقدم معلومات صحية عامة مستخرجة من قاعدة البيانات الموثوقة مع روابط مرجعية دقيقة (Citations).

قواعد التصعيد (Escalation): إذا اكتشف أعراض خطيرة (مثل آلام حادة في الصدر أو ضيق تنفس)، يرفض الإجابة العادية ويظهر فوراً زر الاتصال بالطوارئ أو التوجيه لخط ساخن بشري.

جدول Prompt Log (لإدراجه في Appendix A الخاص بكم)

Tool	Exact Prompt	What it gave	What we changed and why
1	Gemini	Act as a UX designer for Track 9 (Health Information Navigator). Design a core interaction flow focusing on safety guardrails, public health dataset retrieval (CDC/WHO), and strict non-diagnostic rules.	A detailed conversational flow with emergency escalation steps and source citation guidelines.	Added a multi-step guided question structure instead of free text to reduce the risk of ambiguous health queries and ensure better safety alignment.

الخطوة التالية لك مع Claude:

ادخل الآن على Claude واطلب منه توليد المفهوم الثاني (Concept Draft 2) بأسلوب مختلف قليلًا (مثلًا: يركز أكثر على واجهة تفاعلية تعتمد على بطاقات المواضيع الشائعة ونظام الأسئلة السريعة Quick Replies بدلًا من التوجيه الخطي)، ثم قارن بين الاثنين وحددوا الاتجاه النهائي الذي ستختارونه لتسليمه في المراجعة (Check-in)!